

Attribution in Scientific Literature: New Benchmark and Methods

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Abstract

In scientific communication, large language models (LLMs) present a promising yet challenging frontier for **automated source citation**. While previous approaches to citation generation have focused on document and paragraph-level analysis, they have been **hampered by citation ambiguity and LLM overgeneralization**. We introduce **REASONS**, a novel dataset designed to address these limitations, featuring **sentence-level annotations** across 12 scientific domains from **arXiv**. Our comprehensive evaluation framework explores two critical citation scenarios: **indirect queries** (matching sentences to paper titles) and **direct queries** (author attribution), both enhanced with **contextual metadata**. We uncover a complex performance landscape through extensive experiments with state-of-the-art models, including GPT-4o, GPT-3.5, DeepSeek, and smaller variants like Perplexity AI (7B). **While top-tier LLMs achieve high pass percentages in sentence attribution, they continue to struggle with unacceptable hallucination rates** – a crucial metric for scientific reliability. Notably, **our metadata-augmented approach significantly reduced hallucination rates** across all tasks, suggesting a promising direction for improvement. We show that retrieval-augmented generation (RAG) with the mistral model delivers robust performance in indirect queries, reducing hallucination rates by 42% across domains while maintaining competitive precision with larger models. However, adversarial testing reveals **persistent challenges in establishing strong contextual connections between paper titles and their abstracts**, highlighting fundamental limitations in current LLM architectures. REASONS serves as a challenging benchmark for the research community, specifically designed to advance the development of reliable and trustworthy LLMs for scientific applications. Our findings illuminate LLMs' current capabilities and limitations

in source attribution and chart a course for future improvements in other critical domains.

Keywords

Attribution, Source Citation, Large language models, Retrieval-augmented generation, Hallucination rate, Pass percentage

ACM Reference Format:

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1 Introduction

Source Attribution is an effective method to establish grounding in LLM to **achieve trust in generative artificial intelligence** (GenAI) [16]. An LLM that can provide source citation through its learned knowledge or with a support database (e.g., retrieval augmented generation (RAG) [34]) can signal its semantic understanding of the user query and its tendency to hallucinate on specific topics [1]. For instance, adding attribution to the *generated text* can **help detect misinformation** with public news reports as LLMs can generate highly persuasive and realistic content for writing research articles and news reports, making it **challenging for users to distinguish between genuine and fabricated** information [33, 40, 41].

Prior research in source attribution falls into one of four categories: (i) **Evaluating** LLMs' Citation Capabilities Using **RAG**: This involves **assessing how well LLMs retrieve and cite** supporting evidence for generated responses. For example, the ALCE benchmark [19] evaluates citation quality on fluency, correctness, and relevance of the citation through question and answer. (ii) **Context Ablation**: This approach uses techniques such as context ablation to **determine whether a citation is necessary or sufficient** for a response. SelfCite [12] exemplifies this by leveraging self-supervised rewards to improve fine-grained, sentence-level citations. (iii) **Intrinsic Source Attribution**: This focuses on training LLMs to **associate pretraining data with unique document IDs and generate these**

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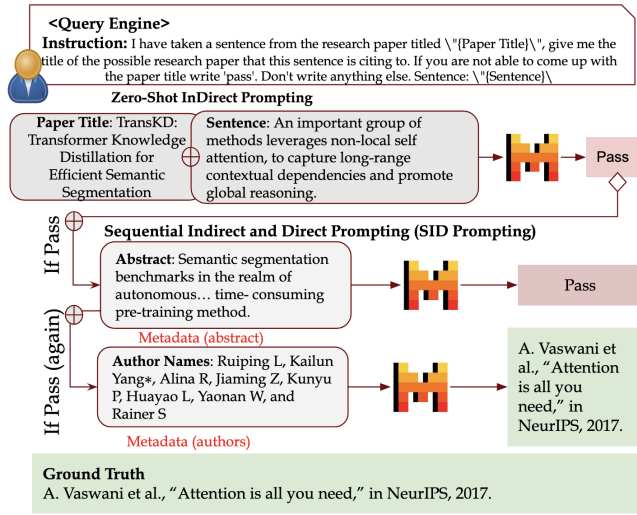


Figure 1: Motivation for REASONS: Improving Citation Generation in LLMs.

IDs as attributions [25, 31]. Source-aware training enables such intrinsic citations, enhancing transparency and verifiability without significantly altering model performance. **These studies has two main limitations: it primarily focuses on general-purpose content rather than specialized domains, and it typically provides attribution at too high a granularity** (with the exception of the recent SelfCite [12]). Using general-purpose domains presents additional challenges, as Wanger et al. demonstrated that **Wikipedia is unsuitable for evaluating LLM attribution capabilities** since much of its content is generated by AI bots [60]. Furthermore, **models trained on paragraph-level or document-level citations tend to overgeneralize or misinterpret user queries** [45, 63]. These approaches fail to meet the requirements of scientific contexts, where precise and detailed attribution is essential.

Figure 1 illustrates a fundamental challenge in automated citation generation: language models often fail to provide accurate references with minimal context. As shown in Figure 1 (top flow), when given only a source paper and query sentence, the model abstains ("Pass"). However, the bottom flow demonstrates our key insight—**progressively adding metadata (abstract followed by author information)** enables successful identification of the correct citation, motivating our research to enhance attribution capabilities for scientific writing and information verification.

(iv) **Resources:** While datasets for training citation-capable LLMs exist, such as **UnarXive** [52] and **S2ORC** [38], they present additional challenges that compound the previously mentioned limitations. These datasets suffer from **incomplete scientific field coverage, data quality issues, and lack sentence-level citation granularity** [36] (see Table 1 for a comparison). Such resource constraints further compromise the accuracy and relevance of model-generated citations, mainly when applied across diverse research disciplines [11].

These challenges extend beyond academic research to commercial applications. Modern commercial search systems powered by LLMs, including Bing Search (which uses GPT-4) [39] and Perplexity AI [51], face similar attribution problems¹. These widely used

¹These commercial tools are accessed via their APIs rather than through their web platforms.

Feature	REASONS	UnarXive	PubMed	CiteULike	S2orc
Sentence-level Annotations	✓	✗	✗	✗	✗
Paper Titles	✓	✗	✓	✓	✓
Abstracts	✓	✓	✓	✓	~
Author Names	✓	✓	✓	✓	✓
Multi-domain	✓	✓	✗	✓	✓
Citation Meta-data	✓	✗	✗	✗	✗
Time Period	2017-2024	1991-2024	1990-2024	2004-2024	2021-02

Table 1: REASONS is uniquely designed as a sentence-level attribution benchmark dataset, while other datasets serve broader purposes like citation recommendation (UnarXive), medical research (PubMed), recommendation systems (CiteULike), or citation and text summarization (S2orc).

tools often fail to properly cite their sources, making it nearly impossible for users to verify whether the information comes from reliable scientific literature or was simply generated by the AI.

To address these challenges and ensure authenticity in source attribution tasks, we present **REASONS**, a new benchmark to examine the source attribution capabilities of LLMs with emphasis on three properties: **(1) high-quality dataset as it is derived from peer-reviewed paper with genuine citations, (2) granular sentence-level citation** and forces LLMs to contextualize on the entities in a sentence, and **(3) rich in semantic as it provides access to metadata for additional context**. **REASONS**, a short form of **REtrieval and Automated citationS Of scieNtific Sentences**, comprises sentences from 12 popular scientific domains on **arXiv**.

Key contributions of this paper are the use of the REASONS dataset to answer the following questions:

- (1) Does an LLM understand the scientific literature? We explore this question using two engineered prompts: **Direct Querying and Direct Querying with Metadata, which are meant for Author Attribution**. These approaches help evaluate LLMs' knowledge awareness regarding scientific content (refer to section 4).
- (2) Does an LLM understand a scientific sentence and correctly identify its sources? This examines the **semantic and conceptual understanding of LLM to generate plausible attributions without actual knowledge of the source materials** (refer to section 4).
- (3) How often do LLMs show defense behavior to avoid hallucination when providing attribution? We introduce **pass percentage (PP)** and **hallucination rate (HR)** as two metrics to evaluate model performance (refer to subsection 5.1).
- (4) We perform the same level of testing on RAG-based LLMs (refer to subsection 4.1), where we examine the **impact of retriever and re-ranker in providing the appropriate citations** while maintaining control over PP and HR.
- (5) To strengthen our findings, we conduct an **adversarial examination to identify semantic-level vulnerabilities and failure modes in LLM source attribution tasks** that remain undetected by standard testing procedures (refer to section 7).

2 Background

As citation systems evolve beyond traditional methods [6, 13, 17, 23, 56], LLMs [9, 48] have emerged as powerful tools for scientific attribution. The progression from rule-based systems to neural approaches has set the foundation for more sophisticated citation generation capabilities. However, challenges remain in ensuring

both accuracy and reliability across scientific domains, particularly when models must determine appropriate sources for specialized content. These developments have led researchers to explore both intrinsic citation capabilities and retrieval-augmented approaches.

Large Language Models in Citation Generation: The advent of LLMs like GPT-3 and its successors has further transformed NLP. Initial language model systems such as those based on BERT have significantly improved citation recommendation by converting unstructured text into meaningful vectors [7, 15, 27]. Recent studies have focused on evaluating the fidelity of generated text to its sources [5, 20, 28, 54, 64]. Rashkin et al. [49] introduced the "attributable to identified sources" (AIS) score, while Bohnet et al. [9] and Honovich et al. [24], Yue et al. [62] have focused on automating AIS. Liu et al. [37] explored human evaluation of commercial generative search engines such as Bing Chat, NeevaAI, Perplexity AI, and YouChat. Byun et al. [11] investigated the accuracy and relevance of LLM-generated citations, finding that while GPT-4 outperformed earlier models on author and title accuracy across different venues, citation relevance remained a challenge. These approaches primarily operate at document or paragraph-level granularity, which limits precision in scientific citation tasks.

Current frameworks also lack adequate metrics to differentiate between partial attributions and complete hallucinations. REASONS fills this gap with sentence-level annotations across scientific domains and introduces specific metrics to measure attribution accuracy and hallucination tendencies. By supporting direct and metadata-augmented queries, REASONS provides a practical benchmark that better aligns with the challenges of evaluating attribution capabilities in specialized (e.g., biomolecules, neurons and cognition) scientific contexts.

Large Language Models in Citation Generation Using RAG: RAG systems improve LLMs by adding citations to generated text, which helps reduce hallucinations. Recent studies show even advanced models like GPT-4 hallucinate about 30% of the time [50]. Researchers have embraced retrieval-augmented LLMs as a promising solution for this problem [10, 18, 22, 26, 30, 32, 53, 61]. Yet these systems face a key challenge: they often either refuse to answer or provide incorrect information. This trade-off has not been thoroughly studied. Our research tests whether RAG-based LLMs can answer all questions and reduce hallucinations. We measure both response PP and HR, developing improved RAG methods that outperform existing systems. We also explore two practical applications: using RAG with smaller, more efficient LLMs and customizing RAG for specific needs [44].

3 REASONS: Source Attribution Dataset

REASONS is a gold-standard dataset comprising sentences **extracted from related work sections of IEEE-formatted papers in Computer Science and Biology published on ArXiv between 2017-2024**. Figure 2 provides a detailed breakdown across **12 scientific domains**, specifying (a) total papers collected, (b) IEEE-formatted papers after filtering, and (c) sentence-level citation counts per domain. We have made the complete dataset and all experimental code available in the GitHub repository². The dataset incorporates **cross-domain citation mapping and categorical classification tags**

²<https://github.com/YashSaxena21/REASONS>

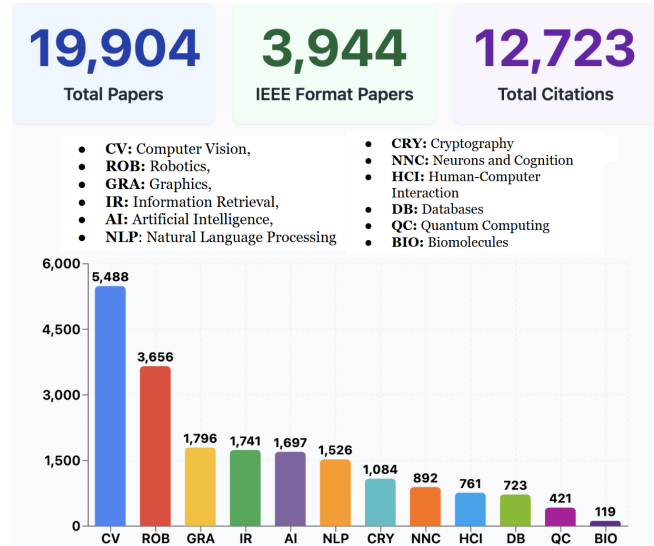


Figure 2: A snapshot of the number of papers in the REASONS dataset by domain, highlighting the coverage of different research areas. The y-axis represents the number of papers, and the x-axis represents all the domains of the REASONS dataset.

REASONS Dataset Sample	
Category	Computer Vision
Link	http://arxiv.org/abs/2012.05435v2
Paper Title	Optimization-Inspired Learning with Architecture Augmentations and Control Mechanisms for Low-Level Vision
Sentence ID	32
Citation Context	
Sentence	We adopt the ℓ_1 -norm. For GM, we establish a residual network with seven convolution layers and six ReLU blocks, which are plugged behind each convolution layer. The DM is constructed as a standard CNN-based classifier.
Citation Information	
Citation Text	C. Ledig, L. Theis, F. Huszar, J. Caballero, A. Cunningham, A. Acosta, A. P. Aitken, A. Tejani, J. Totz, Z. Wang, and W. Shi. "Photo-realistic single image super-resolution using a generative adversarial network." in CVPR, 2017, pp. 105–114.
Cited Paper ID	arXiv:1609.04802
Cited Paper Title	Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network
Cited Paper Metadata	
Cited Paper Abstract	Despite the breakthroughs in accuracy and speed of single-image super-resolution using faster and deeper convolutional neural networks...
Cited Paper Authors	Christian Ledig, Lucas Theis, Ferenc Huszar, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, Wenzhe Shi

Table 2: An example from the REASONS dataset containing a reference to a paper, text of the paper, and details about the cited paper.

to enable fine-grained analysis of **how research propagates across disciplinary boundaries**. We focused on the *related work* section since its citations serve a unique purpose: **establishing research context and highlighting the paper's novel contributions through comparison with existing literature** [59].

Process of Creating REASONS: Our dataset creation pipeline employs a multi-stage process where candidate papers undergo semantic parsing to **isolate related work** sections. Thereafter, **extract individual sentences containing citations** and perform **metadata enrichment** through Oxylabs³ SERP Scraper API. The citation verification protocol requires exact matching between in-text citation

³<https://oxylabs.io/>

strings and search results to ensure metadata accuracy. Each sentence is stored with structured JSON metadata, including source paper identifier, publication date, cited paper metadata including title, authors, venue, and year, additional context with surrounding sentences, classification tag, and cross-domain citation markers. Each field of the JSON is important for following:

(1) *Source paper identifier and publication date*: The source paper identifier helps verify the credibility of the dataset and citation claims while reducing the risk of incorrect attribution. The publication date provides LLMs with context about the recency of the citation and aids in evaluating LLMs based on their knowledge cutoff date.

(2) *Citation target metadata* (title, abstract, authors, venue, year): The citation target metadata helps verify the citation and establish additional context and semantic relevance with the sentence rather than relying solely on keywords.

(3) *Cross-domain citation markers*: Helps in analyzing how research being conducted in one domain connects it with other domains, thereby providing a broader perspective.

Direct Query Prompt: Author Attribution

Task: Extract author information from academic paper

Input: Paper title: "Research Paper Title"

Output format: Structured data

Rules:

- Return a structured data array containing only author names
- Format each name as "FirstName LastName"
- Return ["pass"] if authors cannot be determined with no additional text

Example output: ["John Smith", "Maria Garcia", "Wei Zhang"]

Direct Querying with Metadata

Task: Extract author information from academic paper provided metadata

Input: Paper title: "Research Paper Title"

Metadata: Abstract: "[Abstract text here]"

Output format: Structured data

Rules:

- Must use only information provided in context
- Must use XML tags to structure response
- Format: <authors>Name1, Name2, Name3</authors>
- Use <authors>pass</authors> if authors cannot be determined
- Names should be in FirstName LastName format
- Return only the XML element with no additional text

Example: <authors>John Smith, Maria Garcia, Wei Zhang</authors>

The dataset's machine-processable format facilitates both direct evaluation (e.g., citation generation given context) and indirect assessment (e.g., citation generation given a scientific sentence) of LLM capabilities. Our standardized benchmark includes evaluation metrics designed to measure attribution accuracy, contextual relevance, and appropriateness of citation placement.

Ethical Considerations: We adhered to Oxylabs Acceptable Use Policy⁴ while respecting arXiv's terms of service regarding automated access. Our collection specifically excluded articles marked with "arxiv.org perpetual, non-exclusive license and CC BY-NC-ND" restrictions. Key ethical safeguards implemented include: **License-Aware Collection:** We crawled articles with CC Zero, CC BY, and CC BY-SA licenses. **Usage Limitations:** REASONS is designed specifically for attribution capability assessment, with metrics (HR, PP, BLEU, F-1) selected to prevent misuse for misinformation generation or copyright infringement. **Process Transparency:** We

⁴<https://oxylabs.io/legal/oxylabs-acceptable-use-policy>

have comprehensively provided our GitHub link for the methodology to enable reproducibility, extension, and verification. **Privacy Protection:** No author contact information was utilized, and all processing received formal IRB approval from our organization.

4 Benchmarking LLMs on REASONS

We evaluate LLMs as responsible citation generators using the REASONS dataset (D) through two tasks: (a) *Direct Query*: LLM generates author names when given a paper title, and (b) *Indirect Query*: LLM generates the cited paper title when given a sentence. The direct query task assesses whether an LLM has memorized domain-specific paper metadata during training [8]. For experimentation, we segment D into D_S and D_M . D_S represents sentences and paper titles for which references are to be generated, while D_M contains the supporting metadata.

Direct Querying as the Author Attribution Task Given a title $t_i \in D_S$, the LLM generates the author names in two scenarios: (a) Direct querying: The LLM receives only t_i as input. (b) Direct querying with metadata: The LLM receives t_i , the ground truth abstract $abs_s \in D_M$, and the correct authors $au_s \in D_M$. In both cases, the LLM's task is to output the {author names}. Below are the prompts for Direct Query with and without metadata.

Indirect Query Prompt

Prompt

I have taken a sentence from the research paper titled "Research Paper Title", give me the research paper that this sentence is citing. If you cannot come up with the research paper, write 'pass'. Don't write anything else.

Instruction

Sentence: non-integer ratios between the spatial dimension sizes of the input and the output to pooling layers.

Response

Citation Paper Title.

SID Prompting

Task: Extract citation information from academic text provided metadata

Input:

- Source Paper title: "Research Paper Title"
- Quoted text: "[SENTENCE]"

Available metadata:

- Abstract: "[ABSTRACT TEXT]"
- Authors: "[AUTHOR NAMES]"

Output format: Structured data

Rules:

- Must use only information provided in context
- Must verify if the sentence cites the paper in the source metadata.
- Return only the complete title of the cited paper if confident
- Return only pass if not confident with no additional text

Example response: Machine Learning Applications in Computer Vision

Alternative example: pass

Indirect Querying as Title Attribution Task Given a sentence $s_i \in D_S$, the LLM is prompted to generate a paper title. For indirect querying with metadata, the LLM receives the following input: $s_i \in D_S$, the ground truth abstract $abs_s \in D_M$, and the authors

$au_s \in D_M$, and the model is prompted to generate the {citation paper title}.

SID Prompting: We implement a sophisticated two-stage citation identification approach that leverages both zero-shot and metadata-enriched prompting strategies. Our method, Sequential Indirect and Direct Prompting (SID Prompting), follows a carefully designed progression:

(A) **Initial Indirect Query Phase:** The system begins with a minimal-context indirect query, challenging the model to identify citations based solely on the quoted text in Figure 1 (B) **Adaptive Metadata Enrichment:** When the initial query results in an uncertain response (marked as “pass”) or incorrect identification, the system automatically escalates to a metadata-enhanced direct query. This second phase strategically provides complete author information, full abstract text, and additional contextual signals.

This iterative, context-escalating approach delivers multiple benefits by significantly reducing uncertainty rates, improving citation accuracy metrics, minimizing hallucination through controlled metadata introduction, and maintaining efficiency by only deploying rich-context queries when necessary. By conditionally introducing metadata only when required, SID Prompting achieves an optimal balance between computational efficiency and citation accuracy, outperforming both standard zero-shot and uniform chain-of-thought implementations.

4.1 Retrieval Augmented Generation (RAG)

Our study works on the following three goals: (a) To assess if RAG can reduce incorrect responses from LLMs. (b) To determine if RAG can minimize hallucinations in LLM outputs. And (c) To evaluate RAG’s consistency across various scientific domains (12 in total) when handling both Direct Query and Indirect Queries.

RAG Formulation: Given a corpus of documents R_M and a sentence $s \in R_S$, the document encoder maps $d \in R_M$ to an embedding $E_\theta(d)$ and the query encoder maps s to an embedding $E_\theta(s)$. The top-k relevant documents for s are retrieved based on the sentence-document embedding similarity, which is computed via dot product: $z(s, d) = \exp(E_\theta(s)^T E_\theta(d))$. We start with a bi-encoder retriever using an embedding model from OpenAI.

The retrieved documents are ranked in two ways, which separates Naïve RAG from Advance RAG. Under the Naïve RAG, we use BM25 relevance scoring to rank the documents, whereas, in Advance RAG, we fine-tune an MPNet [55] 12 layer cross-encoder on REASONS document index R_M to better align it with our task of attribution with LLM. For the fine-tuning of the cross-encoder, we use localized contrastive loss (LCL) for two reasons: (a) In R_M , we do not have labeled positive and negative documents, and (b) for a sentence s there is a possibility for more than one true positive documents [47]. LCL is formally defined as follows:

$$\mathcal{L}_{LCL_s} = -\log \frac{\exp(z_{s, \{d^+\}})}{\sum_{d \in G_s} \exp(z_{s, d})}; \quad \mathcal{L}_{LCL} = \frac{1}{|S|} \sum_{s \in R_S, G_s \in R_M^s} \mathcal{L}_{LCL_s}$$

where G_s represents a set of documents for a sentence s , which consist of a set of relevant documents ($\{d^+\}$) and $n-1$ non-relevant documents $\{d^-\}$ sampled from R_M^s using biencoder. The training of LLMs (Mistral and LLAMA) with Advance RAG happens through the standard cross entropy (CE) loss:

$\mathcal{L}_{CE}(\hat{c}|s, \phi) = \sum_{i=1}^b \mathbb{I}(\hat{c}_i^w = c_i^w) \cdot \log \Pr(\hat{c}_i^w | \phi)$, where, ϕ is parameter of the generator LLM and b is the mini-batch fine-tuning in Advance RAG. $\hat{c}_i$ represents i^{th} citation generation, and $\mathbb{I}(\hat{c}_i^w = c_i^w)$ represents word level comparison with ground truth citation.

5 Experimental Setup

In our comprehensive evaluation of source attribution capabilities, we selected a diverse range of LLMs to benchmark against the REASONS dataset. This carefully curated selection includes proprietary and open-source models spanning various architectures, parameter sizes, and retrieval augmentation techniques.

We begin with OpenAI’s models—**o1**, **GPT-4o**, and **GPT-3.5-Turbo**—which require REASONS evaluation because they are widely used and set attribution standards across the AI industry [43]. **Perplexity AI (pplx-7b-chat)** functions as an AI-powered answer engine that explicitly presents information with citation, making REASONS testing crucial for understanding how its attribution practices directly impact user trust in AI-provided knowledge [3]. For open-source models, we evaluate **Mistral (mistral-7b-v0.2-instruct)**, Apache-licensed AI with support for 32,000-token context windows—a feature crucial for source attribution tasks that may require analyzing extensive documents to identify authorship patterns [29] accurately. We also assess **LLAMA 3.1 (llama-3.1-8b-chat)**, Meta’s premier open-source contribution that supports 128,000-token context windows, enabling comprehensive document analysis for attribution [57, 58]. **DeepSeek R-1** model supports a 128,000-token context window, which is sufficiently large compared to smaller size LLMs, because of which we do not apply any RAG strategies to this model [14].

To evaluate RAG approaches, we test Mistral and Llama models with naive and advanced RAG implementations. **Mistral + Naive RAG** and **LLAMA 3.1 + Naive RAG** combine their respective base models with straightforward retrieval techniques, allowing REASONS benchmarking to reveal whether basic information augmentation meaningfully improves source crediting [35]. We contrast these with **Mistral + Advanced RAG** and **LLAMA 3.1 + Advanced RAG**, which implement sophisticated retrieval mechanisms including specialized cross-encoders. REASONS evaluation of these advanced implementations measures whether complex augmentation techniques deliver attribution improvements worth their computational cost and whether refined open-source approaches can achieve proprietary-level citation performance [21].

5.1 Evaluation Metrics

Our evaluation uses four key metrics:

- (1) **BLEU-4 Score** looks at matches of up to 4-word sequences between generated attributions and ground truth.
- (2) **F-1 Score** evaluates the balance between precision and recall, reflecting the models’ effectiveness in capturing key information.
- (3) **Hallucination Rate (HR)** quantifies the model’s tendency to generate incorrect or partially correct citations, revealing its propensity for fabricating information.

$$HR = \frac{1}{2} \left(\frac{1}{Q_D} \sum \mathbb{I}[\hat{c} \neq c] + \frac{1}{|U_w|} \sum_{w=1}^{|U_w|} \mathbb{I}[\hat{c}_w \neq c_w] \right)$$

where Q_D : queries within a domain, and $|U_w|$: total number of unique words in generated citation ($\hat{c}$) and true citation (c). We introduce two sub-scores to capture different facets of the model’s factual reliability: one addresses overtly incorrect or fabricated

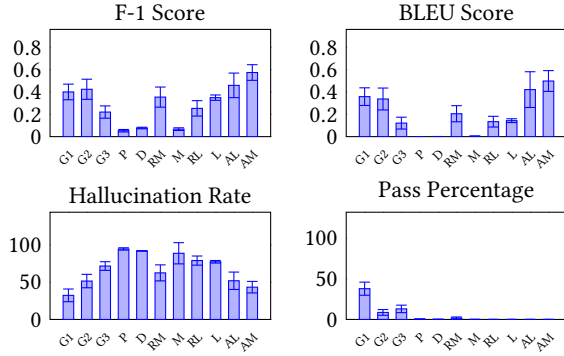


Figure 3: Averaged Zero-Shot Direct Prompting results of different LLMs across all 12 domains. *G1*: o1, *G2*: gpt-4o, *G3*: gpt-3.5-turbo, *P*: pplx-7b-chat, *D*: DeepSeek-R1, *RM*: Naïve RAG mistral-7b-v0.2-instruct, *M*: mistral-7b-v0.2-instruct, *RL*: Naïve RAG llama-3.1-8b-chat, *L*: llama-3.1-8b-chat, *AL*: Advance RAG llama-3.1-8b-chat, *AM*: Advance RAG mistral-7b-v0.2-instruct. For the purposes of clarity and saving space, the terms *AL* and *AM* are used in the figures to denote Advance RAG llama-3.1-8b-chat and Advance RAG mistral-7b-v0.2-instruct, respectively. In the main text of the paper, these are referred to as *AdvRAG(L)* and *AdvRAG(M)*

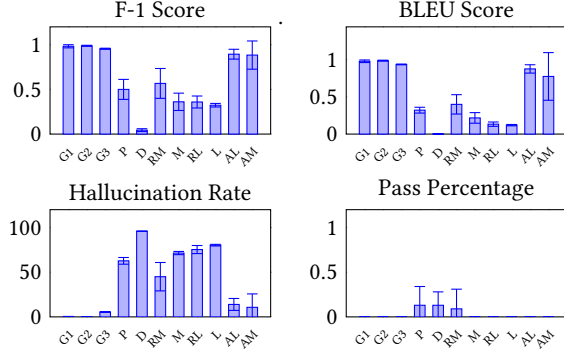


Figure 4: Averaged Direct Prompting with Metadata results of different LLMs across all 12 domains.

information, and the other measures subtler inaccuracies such as minor word level mismatches. Because each sub score can range from 0 to 1, their sum can theoretically reach 2. Multiplying by 1/2 rescales the total so that the overall HR remains within the intuitive [0,1] range. This design preserves interpretability as a fraction while offering a balanced, comprehensive view of the model’s tendency to hallucinate, covering both complete and partial factual errors. This metric is less penalizing than the simple hallucination index proposed in [2].

(4) Pass Percentage (PP) measures the model’s discretion in responding, showing its ability to abstain when uncertain. It is calculated as $\frac{1}{Q_D} \sum I[\hat{c} = \text{Pass}]$. It is important to note that while a high PP can prevent hallucinations by reducing incorrect responses, it may also limit the model’s overall engagement. Furthermore, even with a high PP, a significant HR among the provided responses indicates that the model struggles to differentiate between correct and incorrect citations when it does choose to respond. This underscores the complex interplay between abstention and accuracy in LLM-driven attribution tasks.

6 Performance Analysis

We evaluated model performance through multiple prompting strategies and domain-specific analyses using four key metrics:

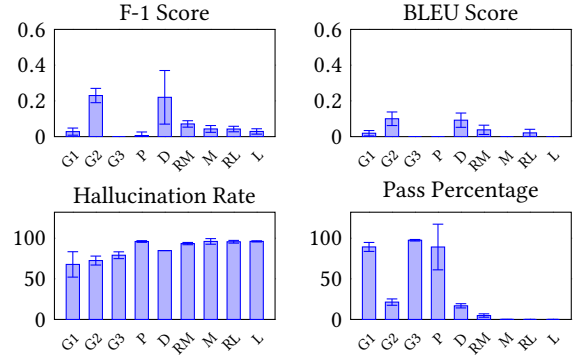


Figure 5: Averaged Zero-Shot Indirect Prompting across 12 domains.

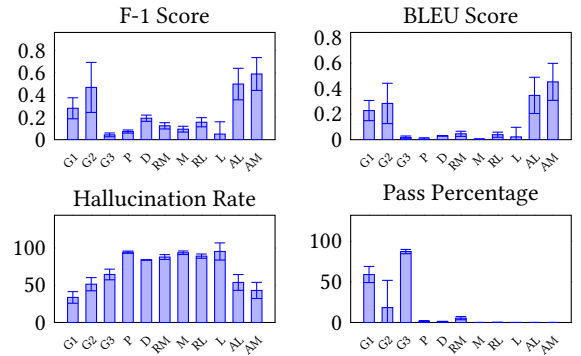


Figure 6: Averaged SID Prompting results of different LLMs across all 12 domains.

HR, which measures incorrect attributions; F-1 Score, which quantifies precision and recall; BLEU Score, which assesses output quality; and PP, which indicates abstention rates. The analysis examines how model performance correlates with domain characteristics, including paper volume, IEEE format representation, and citation patterns, providing contextual interpretation for the observed performance disparities across prompting strategies.

Model Performance on Direct Querying: As shown in Figure 3 zero-shot direct prompting results reveal significant performance variations, with *G1* achieving the lowest HRs (32.3%) and highest F-1 scores (0.40) across domains. Models *P* and *D* consistently underperform with HRs exceeding 94% and minimal abstention (0.34% PP), indicating fundamental limitations in scholarly knowledge representation. Performance improves dramatically with metadata inclusion, reducing *G1*’s HR to 0.4% and *G3*’s from 71.7% to 5.5% (refer Figure 4), demonstrating that these models possess reasoning capabilities but are constrained by information availability in zero-shot scenarios. AdvRAG models significantly outperform the baseline and proprietary models. AdvRAG(L) shows ~30% higher F1/BLEU scores than *G1*-*G3* and AdvRAG(M), achieving the highest overall scores (~60% F1 versus ~40% for baseline models), while both maintain lower HRs compared to most other models.

Effect of Domain Representation on Model Performance: Domain representation (as shown in Figure 7) significantly impacts attribution accuracy, with QC (smallest representation) showing the highest HRs (53.0% for *G1*) as per Table 6 in Appendix B. In contrast, CV (largest representation at 5,488 papers) shows more consistent performance across models. The improvement gap when

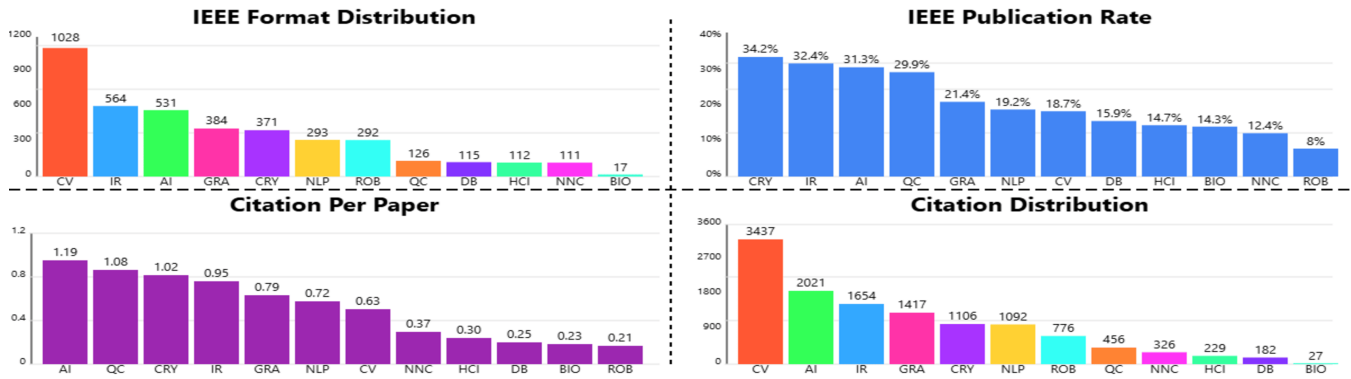


Figure 7: Statistics of the REASONS dataset to understand the source attribution behavior of LLMs.

adding metadata varies inversely with domain size – QC shows 57.8% average improvement compared to CV’s 37.4%. This pattern suggests models develop stronger parametric knowledge for well-represented domains but can compensate for knowledge gaps in specialized domains when provided additional context.

Model Performance on Direct Querying with Metadata: Compared to Figure 3, performance of all models (except D) in Figure 4 improves dramatically across all metrics with metadata enrichment. G1 and G2 achieve near-perfect accuracy with HRs of 0.39% and 0.13%, respectively, while maintaining F-1 scores of 0.98. G3 shows substantial improvement (5.46% HR) compared to zero-shot conditions but lags behind G1/G2. AdvRAG(L) and AdvRAG(M) still perform well (~0.9 F1) but with slightly higher HRs (15-20%) compared to G1-G3’s superior performance in this metadata-enhanced prompting scenario.

Performance of model D declines even further with HR of ~98%, due to its small context window. Notably, all models except P, D, and RM demonstrate 0% PP (see Figure 4), indicating complete confidence in attribution when provided with sufficient context.

Domain-Specific Representation and Performance Effect: Domain characteristics significantly influence metadata-assisted performance. As per Table 7 in Appendix B, BIO show extreme variance, with G2 achieving 0.01% HR while RM struggles at 94.5% HR. NLP consistently presents challenges (highest G1 HR: 0.64%). At the same time, databases show the strongest performance (lowest G1 HR: 0.20%). The AdvRAG variants demonstrate domain-specific optimization benefits, with AdvRAG(M) achieving just 0.07% HR for BIO but struggling with NNC (57.95% HR). These patterns reveal that the effectiveness of metadata varies by domain, with standardized-format domains (CV, Databases) showing consistent improvements than specialized domains with irregular citation patterns.

The results from Figures 3 and 4 demonstrate that metadata provision creates performance convergence among proprietary models (G1-G3) while maintaining a significant gap with RAG models. Interestingly, embedding academic metadata handling capabilities directly in model architecture offers substantial advantages over retrieval-based approaches. Is it true? Direct querying is a test of LLMs’ awareness of scientific knowledge, which it showed with sound confidence. We tested this knowledge in a more realistic scenario by generating a cited paper title from a given sentence using **Indirect querying**.

Model Performance on Indirect Querying: As depicted in Figure 5 and Figure 3, indirect prompting results show substantially

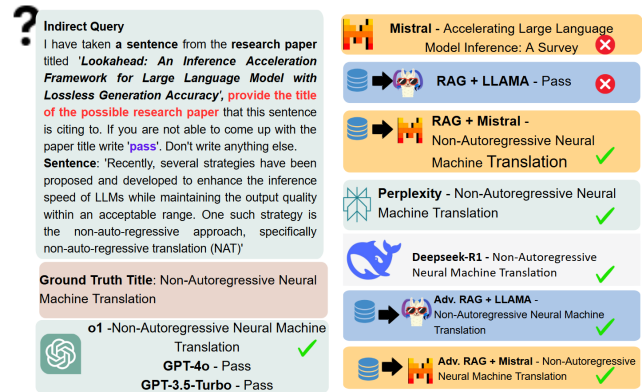


Figure 8: Example of an indirect query where a sentence from a research paper is provided, and the correct title is requested.

higher HRs across all models, with even G1 reaching 67.7% HR compared to 32.3% with direct prompting. G2 maintains relatively better performance (72.5% HR, 0.23 F-1), while G3 demonstrates near-complete failure (79.1% HR, 0.00 F-1). Most notably, G1 shows dramatically increased abstention rates (89.3% PP) compared to direct querying (37.9%), indicating strong uncertainty calibration – G1 “*knows when it does not know*.” Surprisingly, D shows a drastic increase in its performance (85% HR, 0.22 F-1, and 0.09 BLEU), due to its unexpected performance in the Database domain (0% HR, 1.0 F-1, and 1.0 BLEU), indicating the model’s inherent memorization issue. In contrast, RAG models and model P show minimal abstention despite equally poor performance, suggesting dangerous overconfidence in incorrect attributions.

Effect of Domain Representation on Model Performance: Domain characteristics significantly influence indirect attribution capacity. As observed in Table 6 in Appendix B, CV shows the strongest G1 performance (51.8% HR), while Biomolecules exhibit the worst (96.8% HR), despite CV’s larger representation. QC presents particular challenges across models (G1: 91.7% HR, G2: 84.9% HR). These variations correlate with citation patterns – CV’s standardized terminology and higher representation enable more robust retrieval, while specialized domains with unique vocabulary create substantial barriers to indirect attribution. Domain-specific performance gaps widen significantly compared to direct querying, revealing the limits of parametric knowledge when explicit information is unavailable.

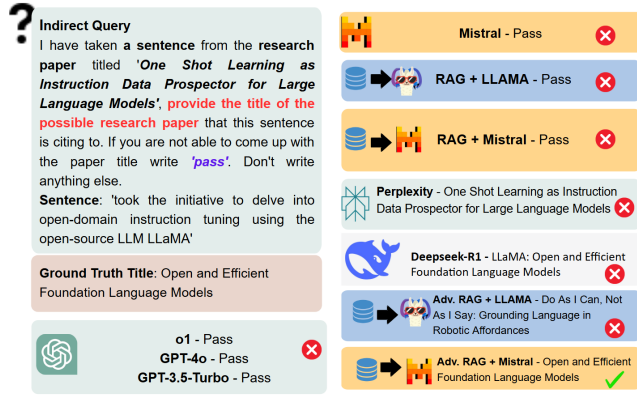


Figure 9: AdvRAG(M) is the only LLM generating the correct source.

Indirect querying exposes fundamental limitations in current attribution systems, with even the best models failing to connect sentences to their sources without explicit metadata reliably. The dramatic gap between direct and indirect performance suggests that current models primarily succeed through information extraction rather than a deep understanding of scientific relationships. Figure 8 and Figure 9 illustrate an indirect query’s structure and the corresponding LLMs’ responses.

Model Performance on the SID Prompting: According to Figure 6, SID results demonstrate intermediate performance between direct and indirect approaches. This hybrid prompting assesses models’ capacity to leverage metadata for attribution verification rather than simple extraction, providing insight into their contextual reasoning abilities. G1 achieves 33.7% HR with a 0.28 F-1 score, substantially better than indirect prompting but worse than direct querying. G2 maintains consistent performance (51.4% HR, 0.46 F-1), while G3 continues to struggle (64.5% HR, 0.04 F-1). PP remains significantly higher than direct querying for proprietary models (G1: 59.1%, G2: 18.3%, G3: 87.3%), indicating appropriate uncertainty calibration. As expected, D shows the worst performance (84% HR, 0.19 F-1, 0.03 BLEU, and 93% PP) due to its context window size. RAG models show near-zero abstention despite poor performance (87.9-96.1% HR), revealing persistent overconfidence issues when verifying attributions. AdvRAG(L) and AdvRAG(M) significantly outperform other models in SID Prompting with F1 scores of ~0.5-0.6 and BLEU scores of ~0.4-0.5 (compared to ~0.2 for G1-G3), while maintaining moderate HRs (~55-45%), much lower than other LLMs except G1.

Domain-Specific Representation and Performance Effect: Referring to Table 9 in Appendix B, domain-specific performance patterns in SID reveal distinct characteristics. Graphics and HCI show the strongest G1 performance (25.5% and 27.4% HR), while QC remains the most challenging (51.8% HR). Notably, domain representation correlates less strongly with SID performance than with direct/indirect approaches, suggesting that verification requires skills different from other prompting styles. Models demonstrate more consistent cross-domain performance in SID than indirect querying, indicating that metadata partially provides sufficient context to overcome domain-specific challenges. However, specialized domains with unique terminology (BIO, QC) continue to present difficulties across all models.

Group	PP(%)	BLEU	F1	HR
Changing Paper Title				
G1	96.23	0.6210	0.8470	17.99
G2	31.45	0.0524	0.2640	83.66
G3	68.55	0.0389	0.1828	87.35
D	89.31	0.0178	0.2318	61.31
RM	3.14	0.0796	0.1584	86.78
RL	5.03	0.0628	0.1448	87.56
AdvRAG(L)	0.00	0.1322	0.4763	85.72
AdvRAG(M)	0.00	0.1569	0.5839	75.41
Changing Paper Abstract				
G1	95.60	0.4595	0.6451	38.49
G2	32.70	0.0396	0.2186	86.22
G3	76.10	0.0034	0.1013	91.64
D	93.08	0.0000	0.0687	94.72
RM	7.55	0.0520	0.1216	89.44
RL	2.52	0.0445	0.1112	90.16
AdvRAG(L)	0.00	0.4101	0.5780	39.67
AdvRAG(M)	0.00	0.4904	0.6954	39.57

Table 3: Performance of LLMs after swapping original titles and abstracts with the most similar ones.

7 Adversarial Experiments

We designed an adversarial experiment to evaluate LLMs’ contextual understanding when attributing sources with modified information. We replace legitimate citation metadata with similar but incorrect alternatives. We use Ratcliff-Obershelp similarity metric (threshold 0.70), we substituted paper titles and abstracts for 200 sentences from the REASONS dataset. Table 3 shows the performance of LLM in adversarial settings. Nearly all models showed high vulnerability to these substitutions. While G1 demonstrated some resilience (96.23% PP for titles, 95.60% for abstracts), most models generated citations based primarily on surface-level similarities rather than genuine contextual comprehension. This vulnerability appeared even in advanced models we expected would perform robustly. The AdvRAG variants showed promise through improved F1 scores despite PP limitations. We found abstract substitutions generally caused more significant performance deterioration than title changes, suggesting deeper semantic understanding remains a significant challenge. Our findings underscore a critical limitation in current LLM architecture for academic applications: the inability to distinguish between legitimate and similar but incorrect sources reliably. To achieve more reliable source attribution, future progress requires integrating knowledge graph [4] representations and graph-theoretic retrieval approaches.

8 Conclusion and Limitations

The REASONS benchmark forms the foundation for developing more trustworthy AI systems for scientific writing assistance, literature review, and knowledge synthesis that appropriately credit original sources. Standardized evaluation across different prompting strategies and domains enables researchers to identify specific attribution weaknesses that must be addressed before deploying AI assistants in high-stakes scientific contexts. Future research need to focus on improving attribution through explicit reasoning mechanisms similar to the Toulmin model within retrieval-augmented frameworks [42]. More sophisticated adversarial testing approaches

including partial abstract modifications and misleading term insertion would provide deeper insights into model robustness.

Our study deliberately excluded mathematics, statistics, and physics papers due to equation prevalence in their related work sections, which the theoremKb crawling method couldn't effectively process [46]. This exclusion allowed us to focus on domains where text-based context more directly influences attribution. We believe that papers from these domains would challenge LLMs as we noticed that current LLMs struggle with mathematical expressions crucial in domains like Quantum Computing, even with RAG.

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A Appendix

A.1 The Story of a Lawyer who employed ChatGPT

In Figure 10, the reliance on LLM-generated content by legal professionals, highlighted by The New York Times, illuminates the pitfalls when these LLMs produce content that lacks proper verification. This incident not only signifies the importance of cross-checking LLM outputs against reliable sources but also exemplifies the potential repercussions of neglecting this critical step. The subsequent requirement for the involved attorney to issue apologies and accept sanctions demonstrates the dire need for robust citation practices in the deployment of LLMs and serves as a crucial learning point for all sectors considering the integration of LLMs into their workflow. Links to the New York Times news articles covering the whole story:

- <https://www.nytimes.com/2023/05/27/nyregion/avianca-airline-lawsuit-chatgpt.html>,
- <https://www.nytimes.com/2023/06/22/nyregion/lawyers-chatgpt-schwartz-loduca.html>
- <https://www.nytimes.com/2023/06/08/nyregion/lawyer-chatgpt-sanctions.html>

A.2 Research Cost Breakdown

The cost associated with this research includes expenses for utilizing OpenAI API, totaling \$640.37. Additionally, the use of Perplexity API incurred costs amounting to \$259.39. Furthermore, GPU resources, we used Replicate⁵ API for our experiments, amounted to \$466.22. For dataset creation, we used Oxylab for \$249 for a month. In total, the expenses for conducting this research sum up to \$1614.98.

A.3 Reproducibility

Our pipeline is straightforward to implement and can be easily reproduced. We have thoroughly documented all experimental details in the main text and the appendices. Although the full text of each prompt is too lengthy to include, we offer examples of each in ?? to help readers understand the style used. *All of our resources, including complete prompt scripts, crawling data, and code for evaluating our approach, are available to the **public repository here**:*

- <https://github.com/YashSaxena21/REASONS>

A.4 Models specifications used during experimentation

The ‘temperature’ hyper-parameter in the LLMs controls the creativity of the LLMs in their response. The lower the temperature, the lower the creativity in the response, and the higher the temperature value, the higher the creativity in the response. By default, the temperature for most of the LLMs is set to 1. The ‘max_tokens’ describes the maximum number of tokens the LLM can generate. The ‘top_p’ is nucleus sampling, which helps limit the irrelevant tokens in the generation.

The ‘top_k’ is the number of retrieved chunks of information that will be considered during the generation in the RAG process.

⁵<https://replicate.com/>

The Story of a Lawyer Who Employed ChatGPT

The New York Times
Artificial Intelligence > A.I. Faces Quiz How the A.I. Race Began Key Figures in the Field One Year of C

Here's What Happens When Your Lawyer Uses ChatGPT

A lawyer representing a man who sued an airline relied on artificial intelligence to help prepare a court filing. It did not go well.

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A lawyer, representing a client against an airline, turned to AI assistance for drafting legal documents. The results were less than ideal.

<https://www.nytimes.com/2023/05/27/nyregion/avianca-airline-lawsuit-chatgpt.html>

ChatGPT Lawyers Are Ordered to Consider Seeking Forgiveness

Steven A. Schwartz and Peter LoDuca must pay a fine and send letters to judges named in a brief filled with fiction, a judge ordered.

Share full article



Legal Consequences for Attorneys Using ChatGPT

<https://www.nytimes.com/2023/06/22/nyregion/lawyers-chatgpt-schwartz-loduca.html>

The New York Times
Intelligence > A.I. Faces Quiz How the A.I. Race Began Key Figures in the Field One Year of

The ChatGPT Lawyer Explains Himself

In a cringe-inducing court hearing, a lawyer who relied on A.I. to craft a motion full of made-up case law said he "did not comprehend" that the chat bot could lead him astray.

Share full article



Lawyer Acknowledges AI Misuse in Court

During court session, an attorney admitted excessively relying on AI, resulting in a legal motion filled with artificial legal references.

<https://www.nytimes.com/2023/06/08/nyregion/lawyer-chatgpt-sanctions.html>

Figure 10: The perils of inadequate verification of LLMs-generated citations in legal documents.

The ‘tokenizer’ converts the retrieved chunks of information and the prompts into tokens.

We have used two different tokenizers ‘*NousResearch/Llama-3.1-8b-chat-hf*’⁶ for LLAMA-3.1-8b-chat and ‘*mistralai/Mistral-7B-v0.2*’⁷ for Mistral-7b-v0.2-instruct. The “Embedding Model” generates embeddings for tokens produced during tokenization. We have utilized the ‘*BAAI/bge-small-en-v1.5*’⁸ model for this purpose. And finally, the Cross-Encoder ‘*ms-marco-MiniLM-L-12-v2*’⁹ is fine-tuned using the LCL function for re-ranking of the retrieved chunks.

Our research utilized a dual-configuration server setup provided by the University. Configuration 1 consists of two nodes, with each node housing 128 cores (totaling 256 cores), 256GB of RAM, and two NVIDIA L40S GPUs, each equipped with 48GB of GPU memory. Configuration 2 is equipped with 8 NVIDIA A100-40GB cards, 1TB of RAM, and 256 CPUs. Due to resource availability in the queue, we alternate between these two configurations. Currently, we have not been able to compare their performance.

We concluded that the Zero Shot Indirect prompting approach is susceptible to hallucinations and is ineffective for the attribution task. Hence, we did not conduct Advance RAG experiments with this prompting due to earlier results from other models, and also, the Advance RAG approach is computationally more expensive Table 6.

⁶<https://huggingface.co/NousResearch/Llama-3.1-8b-chat-hf>

⁷<https://huggingface.co/mistralai/Mistral-7B-v0.2>

⁸<https://huggingface.co/BAAI/bge-small-en-v1.5>

⁹<https://huggingface.co/cross-encoder/ms-marco-MiniLM-L-12-v2>

Hyperparameter	Value
temperature	1.0
max_tokens	256
top_p	0.95
Naïve RAG	
top_k	2
Embedding Model	BAAI/bge-small-en-v1.5
Advance RAG	
top_k	40
Cross-Encoder	ms-marco-MiniLM-L-12-v2
LLAMA-3.1 Tokenizer	NousResearch/Llama-3.1-8b-chat-hf
Mistral Tokenizer	mistralai/Mistral-7B-v0.2

Table 4: Hyper-parameters along with their values used during experimentation

A.5 Dataset Comparison

We contrast the **REASONS** dataset with other similar datasets that could have been utilized for attribution. However, due to constraints within these datasets—such as the absence of sentence-level annotation of citations, metadata of citations, and paper titles—we would not be able to effectively assess the ability of LLMs and RAG LLMs to accurately grasp the context and generate suitable citations (see Table ??). Acronyms used in the paper: Computer Vision (CV), Information Retrieval (IR), Artificial Intelligence (AI) Natural

Language Processing (NLP), Cryptography (Crypto), Neurons and Cognition (NNC), Human-Computer Interaction (HCI), Quantum Computing (QC), and Biomolecules.

A.6 GPU Machine Hours

With the exception of direct prompting, all other prompting styles required a substantial number of GPU hours (see Table 5). Training Advance RAG proved to be a highly time-intensive endeavor, which we attempted to mitigate by alternating between NVIDIA L40S and A100. We also found that LLAMA 3.1 required less time in training than Mistral. The reasons behind this can be a subject of future work. We provide machine-hour estimates to assist other researchers interested in RAG and its applications in provenance and context comprehension, facilitating better time management.

B Individual Results of all the domains across all the prompting styles

A comparative analysis of hallucination rates (HR) across several LLMs in **zero-shot indirect prompting** reveals distinct patterns, focusing on common domains. The **G1, G2, G3, P, RM, M, RL, and L** models consistently show variations in HR. High HR domains like **NNC, Cryptography**, and **NLP** appear recurrently across several models.

Low HR results frequently occur in **IR, CV, and HCI**, indicating a general resilience in these areas across different settings. For instance, **NNC** features prominently with high HR in the **G1, G2, G3, RM, and RL** models, while **IR** and **CV** consistently show low HR across **G1, G2, RM, and M** models.

For **direct prompting with metadata** also shows common domains across the models. Notable high HR domains such as **NNC, IR, NLP, QC**, and **Graphics** feature prominently across different models, indicating frequent challenges in these areas.

Low HR results consistently appear in **CV, NLP, Cryptography**, and **Biomolecules**, showcasing general robustness against hallucinations in these domains. Specifically, **NNC** is recurrently observed with high HR in the **G1, AdvRAG(L), and AdvRAG(M)** models, while **QC** shows up frequently in high HR scenarios (**G1, G2, L, AdvRAG(M)**).

Similarly, **IR** is highlighted in high HR for the **P, RM, RL, and AdvRAG(L)** models, indicating its susceptibility, whereas **NLP** and **Graphics** show variability in HR across multiple models.

For **zero-shot direct prompting** also show significant patterns in common domains.

High HR is commonly observed in domains like **QC, Cryptography, Robotics**, and **Databases**, indicating areas prone to hallucinations. Low HR domains frequently include **IR, HCI, CV, and Biomolecules**, highlighting resilience in these areas.

Specifically, **QC** appears as a high HR domain in the **G1, G2, G3, RL, L, AdvRAG(L), and AdvRAG(M)** models, reflecting a consistent challenge across these models. **IR** and **HCI** are notably present as low HR domains in **G2, G3, AdvRAG(L)**, showing widespread reliability.

Moreover, **Robotics** and **Cryptography** are frequently observed in high HR scenarios in models like **G2, M, and AdvRAG(M)**, while **CV** and **Biomolecules** commonly appear in low HR settings across **G2, G3, M, and AdvRAG(M)**.

For **SID prompting**, high HR domains such as **QC, Cryptography, Databases, NNC**, and **Robotics** frequently appear across several models, highlighting a general susceptibility in these areas. On the other hand, low HR domains commonly include **IR, HCI, CV, and Graphics**, demonstrating resilience against hallucinations.

Specifically, **QC** is observed as a high HR domain in the **G1, G2, G3, RM, RL, AdvRAG(L), and AdvRAG(M)** models, signifying a consistent challenge in this area. **IR** and **HCI** are notably present as low HR domains in **G1, G2, G3, RM, and AdvRAG(L)**, indicating widespread reliability in these areas.

Moreover, **Cryptography** and **Robotics** are frequently observed in high HR scenarios in models like **G1, G2, and RM**, while **CV** and **Graphics** commonly appear in low HR settings across **G2, L, and AdvRAG(L)**. To summarize our results

- The **zero-shot indirect** and **SID** promoting styles are more prone to hallucinations, which lack contextual understanding.
- Notably, **NNC** and **QC** consistently show high HR across multiple models and promoting styles, indicating common challenging domains.
- Conversely, **CV** and **IR** low HR, which show robustness in models, suggesting reliability in these domains across different prompting strategies.

B.1 Further Discussion on Adversarial Examination

This analysis emphasizes the strengths and weaknesses of current LLMs and the need for domain-specific training. It shows that a general approach is insufficient and highlights the importance of specialized training to meet the unique demands of different fields. As LLMs evolve, aligning their development with human knowledge's varied and intricate nature is crucial.

The study finds a significant relationship between the specificity of prompts, especially those with metadata, and the linguistic accuracy of LLMs, as evidenced by higher F-1 and BLEU scores. This suggests that providing detailed, context-rich prompts can significantly improve the quality of generated citations.

Pass Percentage (PP): The varying PP among different models points to a key challenge in LLM development: the ability to understand and reason through complex situations. Models with lower PP struggle with generating relevant responses in complex or critical scenarios, underlining the importance of enhancing reasoning capabilities in LLMs for effective application.

Prompt Design: There is a noticeable difference in how individual models, such as o1 and gpt-4o, respond to different prompts. This underscoring the significance of prompt design in leveraging the full potential of LLMs suggests a complex interplay between the model's structure, prompt formulation, and performance.

Domain	OpenAI -All Models	M	L	D	RM	RL	P	AdvRAG(L)	AdvRAG(M)
AI	34:25	26:03	11:10	34:11	74:49	73:09	34:31	156:24	163:28
Biomolecules	01:11	00:41	00:10	2:31	4:38	4:10	00:20	7:29	7:40
CV	47:45	18:35	19:24	50:22	189:20	198:45	42:05	259:32	302:14
Cryptography	03:50	02:18	04:59	32:21	83:28	89:21	13:23	190:19	194:25
Databases	01:27	00:51	00:40	24:43	49:34	45:46	00:51	96:19	97:48
Graphics	07:08	08:55	06:08	58:43	108:08	127:48	16:52	214:25	227:23
HCI	03:01	01:10	00:42	21:56	48:32	50:51	02:47	95:56	98:44
IR	20:31	11:40	06:52	33:34	91:30	99:43	19:50	193:37	202:23
NLP	28:26	11:42	05:09	47:24	91:07	88:40	13:06	175:58	156:49
NNC	05:00	01:39	02:12	11:29	34:56	41:09	01:19	70:17	84:07
QC	07:26	02:46	01:59	29:07	61:09	67:56	03:17	109:21	113:54
Robotics	19:39	05:41	06:11	22:54	42:07	46:55	09:17	93:07	98:45

Table 5: Time taken by different models with respect to each domain during experimentation, converted to hours and minutes. Red Color: Time recorded while using Replicate API, and Blue Color: Time recording while using NVIDIA RTX 4060 hyperion server.

Zero-Shot Indirect									
Domain	G1	G2	G3	P	D	RM	M	RL	L
Hallucination Rate (%)									
AI	63.61	72.44	81.87	96.27	85.82	93.98	97.16	92.21	95.87
Biomolecules	96.82	69.77	84.68	95.06	86.61	96.63	85.14	96.25	95.57
Crypto	75.04	70.21	81.97	94.16	87.33	93.07	96.11	93.83	97.23
CV	51.83	64.3	79.34	94.63	84.49	91.42	97.12	94.68	95.96
Databases	76.66	69.99	78.93	96.99	0.00	93.42	97.28	95.68	95.84
Graphics	57.49	70.76	85.39	97.25	83.42	92.32	97.55	96.1	95.92
HCI	51.83	73.46	73.41	96.71	84.59	93.01	96.83	96.85	95.61
IR	51.78	67.89	73.41	96.80	85.26	92.01	96.81	96.85	96.01
NLP	63.03	73.98	74.77	97.11	84.79	94.10	97.05	94.29	97.93
NNC	77.27	80.75	82.11	95.49	84.79	94.32	97.13	97.92	96.14
QC	91.72	84.85	76.09	95.15	89.38	92.13	97.14	95.34	95.56
Robotics	55.78	71.55	76.73	95.81	85.50	94.26	97.2	97.51	95.67
Mean	67.73	72.49	79.05	95.95	84.79	93.38	96.04	95.64	96.10
Standard Deviation	15.64	5.51	4.19	1.05	2.52	1.40	3.45	1.67	0.72
F-1 Score									
AI	0.02	0.22	0.00	0.00	0.1564	0.10	0.08	0.07	0.05
Biomolecules	0.00	0.26	0.00	0.07	0.1456	0.09	0.06	0.06	0.05
Crypto	0.01	0.25	0.00	0.00	0.1442	0.08	0.04	0.06	0.04
CV	0.06	0.29	0.00	0.00	0.1714	0.07	0.05	0.05	0.04
Databases	0.00	0.26	0.00	0.00	1.0000	0.09	0.06	0.05	0.04
Graphics	0.06	0.25	0.00	0.00	0.1669	0.05	0.03	0.03	0.01
HCI	0.04	0.23	0.00	0.00	0.1709	0.07	0.03	0.04	0.03
IR	0.06	0.29	0.00	0.00	0.1670	0.04	0.01	0.03	0.02
NLP	0.02	0.21	0.00	0.00	0.1578	0.07	0.04	0.04	0.03
NNC	0.02	0.16	0.00	0.00	0.1578	0.06	0.04	0.02	0.01
QC	0.01	0.13	0.00	0.00	0.1176	0.05	0.02	0.03	0.01
Robotics	0.03	0.21	0.00	0.00	0.1637	0.08	0.05	0.03	0.02
Mean	0.02	0.23	0.00	0.00	0.22	0.07	0.04	0.04	0.02
Standard Deviation	0.02	0.04	0.15	0.02	0.0158	0.01	0.01	0.01	0.01
BLEU Score									
AI	0.01	0.09	0.00	0.00	0.0084	0.05	0.00	0.06	0.00
Biomolecules	0.00	0.12	0.00	0.00	0.0072	0.00	0.00	0.04	0.00
Crypto	0.01	0.12	0.00	0.00	0.0050	0.07	0.00	0.05	0.00
CV	0.04	0.16	0.00	0.00	0.0180	0.02	0.00	0.03	0.00
Databases	0.00	0.12	0.00	0.00	1.0000	0.08	0.00	0.03	0.00
Graphics	0.04	0.12	0.00	0.00	0.0088	0.03	0.00	0.01	0.00
HCI	0.03	0.09	0.00	0.00	0.0177	0.05	0.00	0.02	0.00
IR	0.04	0.14	0.00	0.00	0.0143	0.01	0.00	0.02	0.00
NLP	0.02	0.09	0.00	0.00	0.0092	0.06	0.00	0.00	0.00
NNC	0.02	0.05	0.00	0.00	0.0092	0.02	0.00	0.00	0.00
QC	0.00	0.02	0.00	0.00	0.0030	0.01	0.00	0.00	0.00
Robotics	0.02	0.08	0.00	0.00	0.0044	0.06	0.00	0.00	0.00
Mean	0.01	0.10	0.00	0.00	0.09	0.03	0.00	0.02	0.00
Standard Deviation	0.01	0.03	0.04	0.00	0.0044	0.02	0.00	0.02	0.00
Pass Percentage (%)									
AI	92.92	24.15	97.08	97.77	17.22	4.95	0.05	0	0
Biomolecules	88.89	19.76	97.81	0	14.81	0	0	0	0
Crypto	92.45	20.47	98.17	99.01	18.35	5.63	0.09	0	0
CV	86.7	23.8	95.66	96.48	18.24	3.84	0	0	0
Databases	97.25	20.11	97.67	97.14	17.03	6.23	0	0	0
Graphics	86.38	19.69	97.32	98.8	13.97	1.34	0	0	0
HCI	90.83	19.21	96.61	98.32	19.65	6.11	0	0	0
IR	87.67	16.69	96.61	97.83	16.26	5.21	0	0	0
NLP	92.4	21.98	97.89	98.53	16.77	6.75	0	0	0
NNC	87.73	20.86	98.16	95.21	16.77	6.39	0	0	0
QC	75	17.76	99.34	95.09	10.53	5.72	0	0	0
Robotics	92.91	31.7	97.68	95.95	19.85	5.73	0	0	0
Mean	89.26	21.34	97.50	89.17	16.77	4.82	0.01	0.00	0.00
Standard Deviation	5.528	3.91	0.94	28.11	2.61	2.10	0.02	0.00	0.00

Table 6: Zero-Shot Indirect

Direct with Metadata											
Domain	G1	G2	G3	P	D	RM	M	RL	L	AdvRAG(L)	AdvRAG(M)
Hallucination Rate (%)											
AI	0.32	0.10	6.04	61.31	95.53	37.6	71.39	72.16	80.90	19.24	7.67
Biomolecules	0.46	0.01	5.29	73.99	94.36	94.5	67.98	87.10	79.15	8.15	0.07
Crypto	0.42	0.05	5.41	61.77	95.64	40.87	71.56	73.18	80.45	6.76	4.15
CV	0.42	0.07	4.9	62.35	94.04	41.60	73.67	74.16	78.93	5.51	2.22
Databases	0.20	0.15	5.05	62.55	97.48	39.60	73.33	75.16	0.79	9.73	7.60
Graphics	0.20	0.15	5.43	62.64	95.86	42.31	71.43	78.21	79.80	11.45	8.10
HCI	0.24	0.26	5.26	60.38	95.59	40.75	73.29	75.45	80.66	17.65	7.04
IR	0.39	0.09	5.26	63.88	96.08	48.98	73.1	79.43	80.98	19.71	7.81
NLP	0.64	0.27	6.20	58.79	96.13	37.44	69.68	71.24	80.17	12.60	5.80
NNC	0.51	0.16	5.82	61.12	94.99	38.73	72.04	75.14	81.31	28.11	57.95
QC	0.54	0.17	4.95	61.97	96.40	38.54	69.34	72.09	81.70	18.19	9.25
Robotics	0.45	0.12	5.98	95.33	61.89	39.01	70.62	71.02	80.34	10.27	3.88
Mean	0.39	0.13	5.46	62.72	95.87	44.99	71.45	75.36	80.28	13.94	10.70
Standard Deviation	0.13	0.07	0.44	3.76	1.10	15.89	1.79	4.52	0.90	6.67	15.01
F-1 Score											
AI	0.99	0.89	0.95	0.69	0.04	0.71	0.36	0.33	0.28	0.84	0.92
Biomolecules	0.97	0.99	0.96	0.36	0.06	0.07	0.07	0.21	0.32	0.96	0.95
Crypto	0.93	0.97	0.96	0.61	0.04	0.60	0.40	0.37	0.31	0.91	0.94
CV	0.98	0.99	0.96	0.39	0.06	0.52	0.38	0.34	0.35	0.98	0.98
Databases	0.99	0.98	0.96	0.42	0.02	0.59	0.34	0.34	0.33	0.92	0.95
Graphics	0.99	0.99	0.96	0.45	0.04	0.64	0.44	0.41	0.32	0.94	0.90
HCI	0.99	0.98	0.96	0.34	0.03	0.58	0.35	0.35	0.34	0.82	0.94
IR	0.99	0.98	0.94	0.52	0.04	0.54	0.39	0.39	0.30	0.84	0.92
NLP	0.99	0.92	0.95	0.53	0.04	0.62	0.42	0.40	0.31	0.86	0.91
NNC	0.99	0.99	0.95	0.51	0.04	0.62	0.41	0.36	0.30	0.92	0.39
QC	0.99	0.99	0.96	0.58	0.03	0.65	0.43	0.33	0.29	0.82	0.86
Robotics	0.99	0.99	0.95	0.63	0.05	0.69	0.35	0.49	0.31	0.92	0.95
Mean	0.98	0.98	0.95	0.50	0.04	0.56	0.36	0.35	0.32	0.89	0.88
Standard Deviation	0.01	0.00	0.00	0.11	0.01	0.16	0.09	0.06	0.02	0.05	0.15
BLEU Score											
AI	0.99	0.99	0.93	0.31	0.00	0.43	0.24	0.11	0.12	0.81	0.92
Biomolecules	0.95	0.99	0.94	0.22	0.00	0.00	0.00	0.07	0.12	0.93	0.02
Crypto	0.95	0.97	0.94	0.33	0.00	0.41	0.24	0.13	0.12	0.93	0.95
CV	0.95	0.99	0.94	0.32	0.00	0.39	0.22	0.13	0.13	0.95	0.96
Databases	0.98	0.99	0.94	0.33	0.00	0.41	0.21	0.13	0.13	0.79	0.86
Graphics	0.99	0.99	0.94	0.33	0.00	0.45	0.24	0.17	0.12	0.91	0.91
HCI	0.99	0.98	0.94	0.33	0.00	0.43	0.22	0.13	0.14	0.91	0.92
IR	0.99	0.99	0.94	0.36	0.00	0.48	0.23	0.16	0.11	0.87	0.92
NLP	0.99	0.99	0.93	0.37	0.00	0.46	0.27	0.12	0.12	0.82	0.91
NNC	0.99	0.99	0.93	0.34	0.00	0.46	0.22	0.12	0.11	0.90	0.17
QC	0.98	0.98	0.93	0.28	0.00	0.38	0.26	0.15	0.11	0.80	0.83
Robotics	0.99	0.99	0.93	0.34	0.00	0.49	0.26	0.18	0.12	0.89	0.94
Mean	0.97	0.98	0.93	0.32	0.00	0.39	0.21	0.13	0.12	0.87	0.77
Standard Deviation	0.01	0.00	0.00	0.03	0.00	0.13	0.07	0.02	0.00	0.05	0.32
Pass Percentage (%)											
AI	0.00	0.00	0.00	0.00	0.10	0.00	0.00	0.00	0.00	0.00	0.00
Biomolecules	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Crypto	0.00	0.00	0.00	0.15	0.45	0.00	0.00	0.00	0.00	0.00	0.00
CV	0.00	0.00	0.00	0.03	0.35	0.00	0.00	0.00	0.00	0.00	0.00
Databases	0.00	0.00	0.00	0.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Graphics	0.00	0.00	0.00	0.33	0.14	0.00	0.00	0.00	0.00	0.00	0.00
HCI	0.00	0.00	0.00	0.24	0.00	0.44	0.00	0.00	0.00	0.00	0.00
IR	0.00	0.00	0.00	0.03	0.11	0.67	0.00	0.00	0.00	0.00	0.00
NLP	0.00	0.00	0.00	0.09	0.14	0.00	0.00	0.00	0.00	0.00	0.00
NNC	0.00	0.00	0.00	0.00	0.13	0.00	0.00	0.00	0.00	0.00	0.00
QC	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Robotics	0.00	0.00	0.00	0.00	0.13	0.00	0.00	0.00	0.00	0.00	0.00
Mean	0.00	0.00	0.00	0.13	0.13	0.09	0.00	0.00	0.00	0.00	0.00
Standard Deviation	0.00	0.00	0.00	0.21	0.15	0.22	0.00	0.00	0.00	0.00	0.00

Table 7: Direct with Metadata

Zero-Shot Direct Prompting											
Domain	G1	G2	G3	P	D	RM	M	RL	L	AdvRAG(L)	AdvRAG(M)
Hallucination Rate (%)											
AI	30.9	53.99	73.13	95.64	91.82	56.45	94.23	72.17	76.85	43.77	34.42
CV	35.9	36.32	61.38	95.84	90.90	58.45	92.84	73.17	76.67	35.38	35.43
NLP	27.51	52.49	72.28	96.18	91.78	63.92	93.89	83.17	75.91	47.95	36.63
IR	24.82	42.55	64.19	95.23	91.86	63.12	91.59	77.38	78.16	42.01	37.93
Databases	37.48	53.33	74.08	95.98	92.74	55.45	93.81	74.17	77.92	58.11	40.23
Graphics	29.3	54.29	73.71	95.67	91.78	52.4	92.99	71.19	75.57	47.41	40.26
HCI	22.92	38.02	64.19	95.01	91.92	62.67	92.64	78.15	76.49	38.51	41.11
Biomolecules	21.01	53.25	73.88	90.83	92.75	94.00	43.84	91.2	79.92	67.56	46.28
NNC	36.05	53.13	72.39	93.37	92.23	63.51	91.18	83.73	78.24	48.51	46.31
Crypto	34.41	54.68	73.01	95.39	91.77	54.45	94.78	76.59	76.44	66.16	50.08
Robotics	34.71	56.62	76.29	93.25	91.91	60.89	94.69	81.99	75.92	59.017	50.65
QC	53.04	70.01	82.26	93.70	93.14	65.07	89.75	85.64	81.24	69.108	60.81
Mean	32.33	51.55	71.73	94.67	92.05	62.53	88.85	79.04	77.44	51.95	43.34
Standard Deviation	8.52	9.02	5.80	1.58	0.52	10.76	14.25	6.14	1.73	11.66	7.75
F-1 Score											
AI	0.42	0.39	0.21	0.04	0.06	0.41	0.06	0.31	0.36	0.46	0.53
Biomolecules	0.37	0.42	0.21	0.08	0.07	0.07	0.05	0.14	0.31	0.29	0.65
Crypto	0.42	0.41	0.22	0.04	0.07	0.43	0.06	0.32	0.36	0.40	0.56
CV	0.42	0.60	0.33	0.05	0.09	0.39	0.07	0.32	0.36	0.62	0.62
Databases	0.40	0.42	0.21	0.05	0.06	0.41	0.06	0.31	0.34	0.42	0.55
Graphics	0.49	0.41	0.22	0.05	0.08	0.44	0.07	0.33	0.38	0.42	0.56
HCI	0.51	0.55	0.29	0.05	0.36	0.08	0.07	0.27	0.36	0.62	0.56
IR	0.51	0.52	0.29	0.05	0.07	0.35	0.08	0.26	0.34	0.57	0.69
NLP	0.39	0.38	0.21	0.04	0.07	0.35	0.06	0.21	0.37	0.52	0.66
NNC	0.39	0.39	0.19	0.06	0.07	0.37	0.08	0.24	0.34	0.48	0.57
QC	0.22	0.25	0.12	0.06	0.06	0.34	0.09	0.18	0.30	0.30	0.40
Robotics	0.35	0.36	0.20	0.06	0.07	0.33	0.05	0.15	0.37	0.41	0.54
Mean	0.40	0.42	0.22	0.05	0.07	0.35	0.06	0.25	0.34	0.45	0.57
Standard Deviation	0.07	0.09	0.05	0.01	0.00	0.09	0.01	0.06	0.02	0.10	0.07
BLEU Score											
AI	0.37	0.31	0.11	0.00	0.00	0.24	0.00	0.17	0.15	0.38	0.49
Biomolecules	0.34	0.33	0.10	0.00	0.00	0.00	0.02	0.04	0.11	0.27	0.60
Crypto	0.37	0.32	0.11	0.00	0.00	0.25	0.00	0.18	0.15	0.26	0.47
CV	0.40	0.52	0.23	0.00	0.00	0.24	0.00	0.16	0.15	0.57	0.58
Databases	0.32	0.33	0.10	0.00	0.00	0.25	0.00	0.18	0.14	0.31	0.42
Graphics	0.44	0.31	0.11	0.00	0.00	0.23	0.00	0.19	0.16	0.70	0.51
HCI	0.46	0.46	0.18	0.00	0.00	0.22	0.00	0.13	0.15	0.64	0.51
IR	0.45	0.44	0.18	0.00	0.00	0.28	0.00	0.17	0.14	0.48	0.62
NLP	0.34	0.32	0.11	0.00	0.00	0.21	0.00	0.12	0.16	0.46	0.51
NNC	0.33	0.28	0.11	0.00	0.00	0.19	0.00	0.10	0.14	0.48	0.57
QC	0.17	0.14	0.02	0.00	0.00	0.17	0.00	0.08	0.11	0.20	0.29
Robotics	0.30	0.28	0.09	0.00	0.00	0.18	0.00	0.09	0.16	0.30	0.41
Mean	0.35	0.33	0.12	0.00	0.00	0.20	0.00	0.13	0.14	0.42	0.49
Standard Deviation	0.07	0.09	0.05	0.00	0.00	0.07	0.00	0.04	0.01	0.16	0.09
Pass Percentage (%)											
AI	37.26	9.70	12.37	0.66	0.10	1.65	0.00	0.00	0.00	0.00	0.00
Biomolecules	51.85	6.77	6.77	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Crypto	33.4	5.43	10.52	0.20	0.18	2.15	0.00	0.00	0.00	0.00	0.00
CV	32.26	3.84	8.67	0.09	0.15	3.12	0.09	0.00	0.00	0.00	0.00
Databases	32.42	6.70	10.59	0.95	0.00	2.49	0.00	0.00	0.00	0.00	0.00
Graphics	28.86	6.49	10.30	0.15	0.07	0.45	0.07	0.00	0.00	0.00	0.00
HCI	31.00	8.30	14.51	0.32	0.00	0.56	0.00	0.00	0.00	0.00	0.00
IR	30.11	6.11	14.51	0.86	0.18	0.87	0.00	0.00	0.00	0.00	0.00
NLP	44.6	15.75	17.03	0.18	0.18	1.76	0.00	0.00	0.00	0.00	0.00
NNC	37.12	13.19	21.47	0.74	0.31	1.53	0.00	0.00	0.00	0.00	0.00
QC	50.22	10.09	19.96	0.00	0.00	1.94	0.00	0.00	0.00	0.00	0.00
Robotics	45.10	11.60	9.02	0.00	0.13	4.54	0.00	0.00	0.00	0.00	0.00
Mean	37.85	8.66	12.97	0.34	0.11	1.75	0.01	0.00	0.00	0.00	0.00
Standard Deviation	8.06	3.50	4.61	0.35	0.09	1.25	0.03	0.00	0.00	0.00	0.00

Table 8: Zero-Shot Direct

SID											
Domain	G1	G2	G3	P	D	RM	M	RL	L	AdvRAG(L)	AdvRAG(M)
Hallucination Rate (%)											
AI	29.44	48.49	61.18	95.08	84.08	85.21	94.18	86.68	98.42	51.47	38.45
Biomolecules	35.71	54.99	66.34	95.79	88.94	96.87	86.32	96.51	99.06	52.15	40.89
Crypto	40.44	48.15	66.48	91.18	85.20	85.28	94.78	86.91	98	53.67	45.77
CV	34.44	38.15	59.77	93.47	81.57	87.65	94.13	89.58	99.56	38.82	39.25
Databases	40.74	62.34	66.00	93.91	84.95	86.66	93.96	86.10	98.67	62.49	43.2
Graphics	25.54	62.34	66.55	95.28	82.61	85.91	94.39	86.41	58.83	59.65	47.72
HCI	27.35	39.58	57.01	94.41	82.87	85.68	93.87	88.15	98.12	30.53	23.39
IR	24.01	41.87	57.01	94.68	82.04	85.61	93.33	88.45	98.57	58.58	40.97
NLP	29.2	50.69	61.68	95.87	84.33	88.46	93.88	89.28	98.64	60.26	37.72
NNC	32.68	57.13	74.64	95.97	82.97	88.01	95.14	89.56	99.34	59.42	64.43
QC	51.83	63.63	80.05	92.10	84.56	89.75	95.49	90.73	98.98	69.18	59.84
Robotics	32.45	49.76	57.27	95.07	83.96	89.46	94.36	90.86	98.27	49.24	34.95
Mean	33.65	51.42	64.49	94.40	84.04	87.87	93.65	89.10	95.371	53.788	43.048
Standard Deviation	7.80	8.85	7.16	1.51	2.01	3.25	2.38	2.85	11.51	10.60	10.84
F-1 Score											
AI	0.30	0.54	0.05	0.09	0.20	0.12	0.11	0.20	0.02	0.50	0.61
Biomolecules	0.15	0.51	0.03	0.05	0.14	0.05	0.03	0.05	0.00	0.52	0.57
Crypto	0.35	0.67	0.03	0.07	0.19	0.13	0.10	0.19	0.02	0.62	0.71
CV	0.35	0.67	0.06	0.09	0.23	0.13	0.11	0.16	0.03	0.72	0.73
Databases	0.21	0.03	0.03	0.08	0.19	0.14	0.10	0.19	0.02	0.29	0.48
Graphics	0.41	0.03	0.03	0.05	0.21	0.13	0.09	0.18	0.41	0.38	0.58
HCI	0.33	0.66	0.07	0.08	0.22	0.15	0.13	0.18	0.03	0.70	0.85
IR	0.38	0.64	0.07	0.08	0.23	0.14	0.12	0.15	0.02	0.43	0.68
NLP	0.30	0.51	0.05	0.07	0.19	0.16	0.10	0.13	0.02	0.41	0.49
NNC	0.21	0.45	0.03	0.09	0.17	0.11	0.08	0.17	0.00	0.50	0.31
QC	0.10	0.37	0.02	0.06	0.18	0.10	0.07	0.13	0.01	0.31	0.42
Robotics	0.28	0.54	0.05	0.07	0.19	0.13	0.09	0.14	0.02	0.60	0.62
Mean	0.28	0.46	0.04	0.07	0.19	0.12	0.09	0.15	0.05	0.49	0.58
Standard Deviation	0.09	0.22	0.01	0.01	0.02	0.02	0.02	0.04	0.11	0.14	0.14
BLEU Score											
AI	0.25	0.31	0.02	0.00		0.06	0.00	0.04	0.00	0.32	0.51
Biomolecules	0.14	0.34	0.01	0.00	0.02	0.00	0.00	0.00	0.00	0.32	0.56
Crypto	0.27	0.48	0.01	0.00	0.02	0.06	0.00	0.06	0.00	0.47	0.55
CV	0.25	0.46	0.03	0.00	0.05	0.03	0.01	0.06	0.00	0.51	0.51
Databases	0.17	0.01	0.01	0.00	0.01	0.06	0.00	0.03	0.00	0.12	0.42
Graphics	0.35	0.01	0.01	0.00	0.03	0.03	0.00	0.01	0.26	0.22	0.44
HCI	0.28	0.45	0.03	0.00	0.04	0.07	0.01	0.05	0.00	0.53	0.71
IR	0.32	0.39	0.03	0.00	0.04	0.07	0.01	0.07	0.00	0.54	0.45
NLP	0.26	0.27	0.03	0.00	0.03	0.04	0.01	0.04	0.00	0.23	0.43
NNC	0.15	0.24	0.01	0.00	0.03	0.05	0.00	0.05	0.00	0.40	0.11
QC	0.08	0.17	0.00	0.00	0.03	0.04	0.00	0.03	0.00	0.20	0.31
Robotics	0.22	0.28	0.03	0.00	0.03	0.04	0.00	0.03	0.00	0.30	0.44
Mean	0.22	0.28	0.01	0.00	0.03	0.04	0.00	0.03	0.02	0.34	0.45
Standard Deviation	0.07	0.15	0.01	0.00	0.01	0.02	0.00	0.02	0.07	0.14	0.14
Pass Percentage (%)											
AI	56.8	4.21	87.14	1.86	0.89	7.25	0.00	0.00	0.00	0.00	0.00
Biomolecules	74.07	7.21	89.98	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Crypto	53.34	3.6	89.7	0.84	1.63	6.89	0.00	0.00	0.00	0.00	0.00
CV	52.3	1.6	83.42	0.79	0.67	4.94	0.00	0.00	0.00	0.00	0.00
Databases	63.19	89.98	90.61	0.00	0.00	6.04	0.00	0.00	0.00	0.00	0.00
Graphics	44.25	88.91	90.19	0.64	0.71	6.29	0.00	0.79	0.00	0.00	0.00
HCI	54.15	0.44	83.68	0.96	1.75	4.37	0.00	0.00	0.00	0.00	0.00
IR	49.52	1.45	83.68	0.79	0.60	4.39	0.00	0.00	0.00	0.00	0.00
NLP	57.33	5.49	86.45	2.38	0.93	4.91	0.00	0.00	0.00	0.00	0.00
NNC	69.33	5.21	87.42	2.88	0.92	5.93	0.00	0.00	0.00	0.00	0.00
QC	76.75	7.46	88.6	2.14	0.93	5.97	0.00	0.00	0.00	0.00	0.00
Robotics	57.6	3.35	86.86	2.65	0.93	7.31	0.00	0.00	0.00	0.00	0.00
Mean	59.05	18.33	87.31	1.30	0.93	5.357	0.00	0.65	0.00	0.00	0.00
Standard Deviation	9.92	33.53	2.63	1.02	0.52	1.97	0.00	0.22	0.00	0.00	0.00

Table 9: SID